

Qwen3.5-9B EN: religion — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	9.11 [5.78, 12.44]				1.89	8.89	0.16	1.93	2.99	69.51	0.60	2.05
SFT (LoRA, merged)	implicit	2.89 [1.33, 4.67] ♦	-6.22 ♦	41 : 13	2e-04	0.42 (1.89) ♦	1.56 (8.89) ♦	0.01 (0.16) ♦	0.72 (1.93) ♦	27.31 (2.99)	81.73 (69.51)	0.65 (0.60)	0.00 (2.05) ♦
dpo	implicit	4.22 [2.44, 6.22]	-4.89	41 : 19	6e-03	0.57 (1.89)	2.00 (8.89)	0.02 (0.16)	1.01 (1.93)	27.09 (2.99)	82.59 (69.51) ♦	0.60 (0.60)	0.00 (2.05) ♦
SFT (LoRA, merged)	implicit	5.33 [2.89, 8.00]	-3.78	37 : 20	0.03	0.64 (1.89)	3.11 (8.89)	0.02 (0.16)	1.33 (1.93)	18.37 (2.99)	81.93 (69.51)	0.65 (0.60)	0.00 (2.05) ♦
dpo	implicit	6.67 [4.22, 9.33]	-2.44	33 : 22	0.18	1.14 (1.89)	5.11 (8.89)	0.05 (0.16)	1.28 (1.93)	13.56 (2.99)	76.86 (69.51)	0.68 (0.60)	0.49 (2.05)
baseline (no mitigation)	explicit	7.11 [4.67, 9.78]				0.66	2.44	0.03	1.51	17.85	80.81	0.68	0.00
SFT (LoRA, merged)	explicit	2.67 [1.11, 4.44] ♦	-4.44 ♦	29 : 9	2e-03	0.23 (0.66) ♦	1.11 (2.44)	0.00 (0.03)	0.57 (1.51) ♦	28.52 (17.85)	83.19 (80.81) ♦	0.66 (0.68)	0.00 (0.00) ♦
SFT (LoRA, merged)	explicit	2.67 [1.11, 4.44] ♦	-4.44 ♦	29 : 9	2e-03	0.30 (0.66)	1.11 (2.44)	0.01 (0.03)	0.64 (1.51)	18.89 (17.85)	81.90 (80.81)	0.65 (0.68)	0.00 (0.00) ♦
dpo	explicit	3.11 [1.56, 4.89]	-4.00	27 : 9	4e-03	0.40 (0.66)	1.78 (2.44)	0.01 (0.03)	0.89 (1.51)	11.33 (17.85)	75.93 (80.81)	0.68 (0.68)	0.00 (0.00) ♦
dpo	explicit	3.11 [1.56, 4.89]	-4.00	30 : 12	8e-03	0.25 (0.66)	0.67 (2.44) ♦	0.00 (0.03) ♦	0.59 (1.51)	29.33 (17.85)	82.86 (80.81)	0.60 (0.68)	0.00 (0.00) ♦

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.